

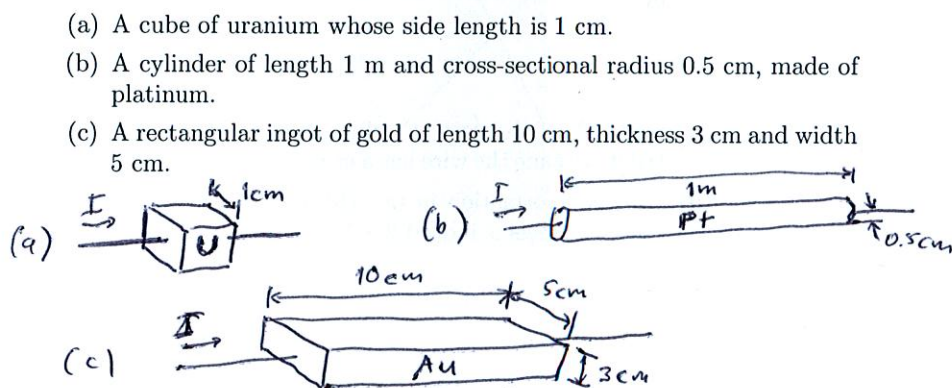
MP204 – Electricity & Magnetism

Problem Set 2

Due in the assignment box by 5pm on Friday, 27 March 2015

In Problems 1 and 2, it is up to you to do some research to find values for the conductances, mass densities, atomic mass numbers, etc. that you'll need to find the answers. And **cite your sources**; points will be deducted if you don't.

1. Three objects are put into a circuit as shown in the pictures; find their resistances.



2. A sample of an element of atomic mass number a has a mass density of ρ . Each atom can contribute k electrons to a current flowing through it.

- (a) Explain why the number density n of conduction electrons in this sample is given by

$$n = \frac{k\rho}{am_p}$$

where m_p is the mass of a proton.

- (b) A thin rod made of indium has cross-sectional area 3 cm^2 . Given that each indium atom has three conduction electrons, use the result in (a) and the formula $\vec{J} = nq\vec{v}$ to determine the drift speed of the conduction electrons in this rod if a current of 150 mA is run through it.
3. The original definition of the basic unit of current, the *ampere*, was as follows: suppose you have two parallel wires, each one metre long, separated by one metre. One ampere is that amount of current such that, when it is run in the same direction through both wires, results in each attracting the other with a force of 2×10^{-7} newtons. We wish to verify this.

- (a) Explain why the two wires *attract* each other. Remember that a charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ feels a force $q\vec{v} \times \vec{B}$. Also, recall that a current flowing in a certain direction is due to electrons moving in the *opposite* direction.
- (b) Although the wires are of finite length, assume that the magnetic field produced by each is that of an infinitely long wire. Use this to find the magnitude (in tesla) of the magnetic field on one wire due to the other.
- (c) Prove the following: if a wire of length L carries a constant current I , then the magnitude of the drift velocity of the electrons is given by

$$|\vec{v}| = \frac{IL}{Ne}$$

where N is the total number of conduction electrons in the wire. (Hint: assume the wire has a cross-sectional area A and use $\vec{J} = qn\vec{v}$.)

- (d) Use the information in (a), (b) and (c) to show that the force that each wire feels is indeed 2×10^{-7} newtons.